Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Shoukas, Artin A. Ph.D.		
Project Title:	Mechanics of Cardiovascular De-Conditioning		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21205-2109	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	02/29/2004
No. of Post Docs:	4	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-CA.002.03		
Performance Goal No.:			
Performance Goal Text:			

Changes in cardiac output result from the altered, myocardial contractility or through changes in venous filling pressure via the Frank Starling mechanism. Our laboratory has previously shown the importance of veno regulation by the carotid sinus baroreceptor reflex system on overall circulatory homeostasis, and in particular the regulation of cardiac output. Decreases in SV responses to an orthostatic challenge are the seminal patho-physiologic observation after space flight. In addition there is evidence that there is a significant degree of cardiac hypertrophy after long during space flight. The exact effects of the hypertrophy on ventricular performance is clearly unknown and could significant contribute to the ortho static intolerance after long term space flight. Our grant aims to test our hypothesis that alterations in venous capacitance function by the carotid sinus baroreceptor reflex system is an important determinant of the cardiac output response seen in astronauts after returning to earth from long term exposure to micro gravity. In addition the physiological effects of cardiac hypertrophy on orthostatic intolerance will be determined. We will use the hind limb unweighted rat model to simulate the patho physiological effects as they relate ` to cardiovascular deconditioning in

Task Description:	micro gravity. To determine mechanisms of impaired stroke volume responses integrated cardiovascular function (in vivo) and contractile reserve will be tested using miniaturized conductance micro manometry catheters. The role of cardiac atrophy in cardiovascular deconditioning will be tested using magnetic resonance imaging to noninvasively measure cardiac mass. Since venous capacitance function and arterial resistance determine ventricular preload and after load respectively, mechanisms of impaired contractile responses in both arterial, venous, and pulmonary vascular beds will be studied. Molecular mechanisms of endothelial dependent (eg. nitric oxide), and independent (Ca2+ homeostasis and vascular smooth muscle myofilament Ca2+ sensitivity) vascular hypo responsiveness to sympathetic stimulation will be studied, using vascular contractility bioassays (in vitro), pressure dimension analysis both in vivo and in vitro, and intracellular Ca2+ measurement (fluorescence spectrophotometry). We plan to test novel, countermeasures based on mechanisms that impair both cardiac output responses and vascular hypo responsiveness in our rat model. These studies will provide important new data concerning normal capacitance vessel function in compensating for postural blood volume redistribution, test our novel hypothesis regarding the pathogenesis of orthostatic intolerance following micro gravity exposure, and provide insights into potential countermeasures and therapies to prevent problematic postural hypotension on reentry. Our laboratory currently performs experiments from chronic instrumented animals to the cellular and molecular mechanisms involved in cardiovascular regulation and control.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will provide important new data concerning normal capacitance vessel function in compensating for postural blood volume redistribution, test our novel hypothesis regarding the pathogenesis of orthostatic intolerance following micro gravity exposure, and provide insights into potential countermeasures and therapies to prevent problematic postural hypotension on reentry. The direct earth benefit in these studies is that orthostatic intolerence is a prevelant problem in the aging population, particular to the female aging population. These data will provide additional insite into the problem and may have a significant role in curing this problem.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 05/05/2011)